

ProjectOne Engineering Handbook

Chapter 5 — Next.js Architecture

5.1 Objective

Next.js provides the application framework for ProjectOne. The architecture must prioritize performance, scalability, SEO where applicable, and a clean separation between server and client responsibilities.

5.2 Architectural Principles

- Server-first architecture.
- Client code only when required.
- Small, reusable modules.
- Predictable routing.
- Clear separation of UI, business logic and data access.

5.3 App Router

Use the App Router as the standard routing system. Route groups should organize features without affecting URLs. Layouts should minimize duplication and maximize consistency.

5.4 Server Components

Server Components are the default choice. Data fetching should occur on the server whenever possible to reduce client-side JavaScript and improve performance.

5.5 Client Components

Only use Client Components when browser APIs, local interactive state, animations or event handlers require them. Minimize the client bundle.

5.6 Data Fetching

Perform data fetching close to the server. Avoid unnecessary client requests. Cache data appropriately and invalidate caches explicitly when data changes.

5.7 Folder Organization

Routes contain only routing concerns. Business logic belongs in services. Shared UI belongs in reusable component libraries. Utility functions remain framework-independent whenever practical.

5.8 Performance

Optimize images, fonts and bundles. Lazy-load expensive components. Measure performance using profiling tools before introducing optimizations.

5.9 Error Handling

Every route should define loading states, error handling and not-found pages where appropriate. Unexpected failures must degrade gracefully.

5.10 Security

Never expose secrets in client code. Validate all server actions, sanitize inputs and enforce authorization on the server regardless of frontend restrictions.

5.11 Anti-Patterns

Avoid:

- unnecessary Client Components
- duplicated layouts
- fetching the same data multiple times
- business logic inside pages
- direct database access from UI components

5.12 Code Review Checklist

Confirm:

- correct Server/Client separation
- reusable layouts
- optimized data fetching
- consistent routing
- secure server actions
- acceptable bundle impact

Chapter Summary

Next.js should provide a fast, scalable and maintainable application architecture. Server-first thinking and modular design are mandatory engineering principles for ProjectOne.